

36 Months. Storage Studio → Complete CCUS Platform.

The full 36-month technical implementation plan: PoC execution, ML model portfolio, module build sequence, security roadmap (SOC 2), academic partnership programme, team hiring plan, and the decision gates that determine when to raise, when to scale, and when to consider strategic acquisition.

36-Month Roadmap Architecture

HOW TO USE THIS DOCUMENT

This roadmap covers the full 36-month journey from PoC to production platform. **Phase 0 (Weeks 1–13)** is the PoC sprint — fully detailed in the separate PoC Implementation Roadmap. This document references Phase 0 outputs and picks up at Phase 1. Each phase has explicit entry criteria (what must be true before starting), deliverables (what gets built), and exit criteria (what must be validated before proceeding).

Month 3 (Week 13)

PoC

Working studio live. 3 trial users. Patent filed.

Month 13-14

\$10K

MRR milestone. Pre-seed raise initiated.

Month 24

Full

Complete CCUS platform. Team of 3. \$100K MRR target.

Month 36

\$1M+

ARR. Decision point: scale or strategic exit.

PHASE	DURATION	FOCUS	ENTRY CRITERIA	REVENUE TARGET
Phase 0 – PoC	Weeks 1–13	Storage Studio PoC: 3D viz, ML wiring, PO/P1 features, patent, deploy	Start now	First 3 paying customers
Phase 1 – Revenue	Months 4–13	ML model completion, full permit suite, Monte Carlo, academic outreach, \$10K MRR	PoC live, ≥1 trial customer	\$10K MRR
Phase 2 – Platform	Months 14–24	Capture/Transport/Utilization modules, MMV, GIS import, team collab, 8 jurisdictions, hire Eng #1	\$10K MRR, pre-seed closed, provisional patent filed	\$50–100K MRR
Phase 3 – Enterprise	Months 25–36	Enterprise security (SOC 2), AI assistant, API layer, hire Eng #2 + Domain Expert, enterprise deals	Full platform live, 2 enterprise contracts	\$100K+ MRR, \$1M+ ARR

Revenue — \$10K MRR

Entry criteria: carbonlens.io is live. At least 1 trial user. Provisional patent filed. **Duration:** Months 4–13 (10 months, 7 hrs/week). **Exit criteria:** \$10K MRR, design-partner customer, pre-seed investor conversations started.

Months 4–6 ML Model Completion + Full Permit Suite

~84 hrs

- ☐ Complete ML model training: GMDH/MGGP/MARS on full IFT dataset (1,200+ points including UTP lab data). Achieve $R^2 > 0.98$. Calibrate bootstrap P10/P50/P90 with uncertainty-toolbox. [CODE](#)
- ☐ Train CO₂-EOR Minimum Miscibility Pressure (MMP) model (XGBoost on SPE literature dataset). This was in v1 but dropped in v2. Restore it — it unlocks EOR operator ICP. [CODE](#)
- ☐ Build full permit suite: MAIP, AoR radius at P10/P50/P90, CO₂ column height, phase state, full property table with citations. PDF export via pdfmake. EPA Class VI format. [CODE](#)
- ☐ Build basic MMV planner (P2 from PoC): monitoring well location calculator (AoR P90 radius), pressure monitoring schedule, USDW delineation from surface to injection zone depth. [CODE](#)
- ☐ Submit academic paper to JPSE or IJGGC. Collect validation data from Sleipner preset comparison. Co-author with UTP supervisor if possible. [WRITE](#)
- ☐ Reach out to 3 target professors (UTP, University of Wyoming, UT Austin Bureau of Economic Geology). Offer free Studio access in exchange for case study using Sleipner or Decatur preset. [BIZ](#)

Months 7–10 Monte Carlo Engine + Site Suitability + SPE Presence

~112 hrs

- ☐ Build Monte Carlo UQ engine (n=200 Professional, n=1,000 Studio) fully in Web Workers. Monte Carlo histogram in sidebar. AoR risk envelope chart in D3.js. [CODE](#)
- ☐ Train Site Suitability Scorer (Random Forest) on 28 real Class VI permitted sites from EPA Data Repository. 0–100 score with sub-scores. Feature importances shown to engineers. [DATA](#)
- ☐ Build NETL SRMS volumetric and mass-based storage capacity estimation. Auto-populated from reservoir builder parameters. Displayed alongside suitability score. [CODE](#)
- ☐ Attend or submit to SPE Annual Technical Conference. Live 3D demo at booth or in paper presentation. Target: 5 business card exchanges with consultancy engineers or project operators. [BIZ](#)
- ☐ Add Studio tier team features MVP: invite a colleague to view a project (read-only). Share project URL. This gates the Studio upsell from Professional. [CODE](#)

Months 11–13 \$10K MRR Sprint + Pre-Seed Groundwork

~84 hrs

- ☐ Outreach sprint: 50 personalised emails to EPA tracker contacts, LinkedIn simulation demo video post (target 10K views), SPE YP webinar application. [BIZ](#)
- ☐ File provisional patent #2: UQ-to-permit translation method (automated AoR from ML confidence intervals). [IP](#)
- ☐ Build investor materials: 10-slide pitch deck, 3-year financial model, product demo video (3 min), one-pager. Target investors: Congruent Ventures, Clean Energy Ventures, Buoyant Ventures. [BIZ](#)

- ☐ Co-author validation paper with academic partner. Submit for peer review. This is the pre-seed fundraise's strongest technical credibility signal. [WRITE](#)
- ☐ **Target: \$10K MRR achieved by end of Month 13.** If not achieved, diagnose conversion failure before adding features. Talk to people who tried the demo and did not convert. [BIZ](#)

Full Platform — \$100K MRR

Entry criteria: \$10K MRR, pre-seed closed (\$300K–\$500K), provisional patent filed, 1 design-partner enterprise customer. **Duration:** Months 14–24. **First hire at Month 14–16: Full-Stack Engineer.**

Months 14–17

Capture + Transport + Utilization Modules & Engineer Hire

Full-Stack Engineer

☐ Hire Full-Stack Engineer (React + Three.js specialist). First task: implement GIS shapefile import — converts real basin geometry to parametric reservoir in the builder. HIRE

☐ Build Capture Module: CO₂ source characterisation inputs (purity, flow rate, impurities). Technology selector (post-combustion, oxy-fuel, DAC). Capital + OPEX estimator benchmarked against IEA/NETL capture cost data. Purity impact feeds storage module IFT calculation. CODE

☐ Build Transport Module: dense-phase CO₂ pipeline sizing (Darcy-Weisbach). Phase behaviour during transport. Pressure drop over distance. Pump/compressor sizing. Impurity effects on transport viscosity. CODE

☐ Build Utilization Module: utilization pathway selector (EOR, mineralisation, food-grade, concrete curing, synthetic fuels). CO₂-EOR MMP predictor (ML model trained in Phase 1). 45Q utilization vs. storage credit comparison calculator. CODE

☐ Rename product from “CarbonLens Storage Studio” to “CarbonLens” with the full CCUS positioning now credible. Update all marketing materials. BIZ

☐ Add 4 new regulatory jurisdiction templates: Australia (NOPSEMA/GHGSRs), Canada (AER Directive 51), EU ETS Annex I, Saudi Arabia (ARAMCO + GAMEP). CODE

Months 18–21

Full MMV Planner + Team Collaboration + Enterprise Security Layer

ML Engineer Hire

☐ Hire ML Engineer (Python, GP, Bayesian methods). First task: upgrade UQ from bootstrap to Gaussian Process confidence intervals (Bell & Harvey 2021 methodology). More rigorous and faster than Monte Carlo. HIRE

☐ Build full MMV Planner: monitoring well location optimiser (constrained by AoR P90 and formation geometry), pressure monitoring schedule generator, USDW protection zone delineation, seismic survey planning (vintage, frequency, coverage). CODE

☐ Build Team Collaboration Workspace: Cloudflare D1 project library, version history (append-only KV), role-based permissions (Owner/Editor/Viewer), client-branded PDF report generation. CODE

☐ Implement enterprise security layer: SAML 2.0 SSO (Okta, Azure AD), data residency options (EU/US/APAC Cloudflare regions), field-level encryption for sensitive formation data. SECURITY

☐ Penetration test (third-party provider). Remediate all critical/high findings. Publish security page on carbonlens.io. Begin SOC 2 evidence collection. SECURITY

☐ File full utility patent #1 (from provisional #1 filed in Week 7 of PoC). Use pre-seed proceeds to engage patent attorney for full prosecution. PCT filing for international protection. IP

- ☐ Activate academic partnership programme: formalise agreements with 2–3 universities. Co-author curriculum materials. Work with UTP to integrate CarbonLens into reservoir engineering for CCS course. ACADEMIC
- ☐ Close first 2 enterprise contracts (\$3K/month minimum each). Use design-partner relationship from Phase 1 as primary enterprise sales lead. Prepare SOC 2 questionnaire response for enterprise procurement. BIZ
- ☐ Add ECLIPSE/CMG output import (view-only): engineers can load competitor simulation results alongside CarbonLens outputs for comparison. This is a Trojan horse for enterprise adoption. CODE
- ☐ Expand formation library to 12+ presets including Weyburn-Midale (Canada), In Salah (Algeria), Quest CarbonSAFE (US Midwest), Malaysia Offshore (PETRONAS CCS). DATA

Enterprise Platform — \$1M+ ARR

Entry criteria: Full platform live, 2 enterprise contracts signed, SOC 2 Type I audit started. **Duration:** Months 25–36.
Hires: Domain Expert (Month 20–24) + consider Series A at Month 30.

Months 25–28 Subsurface AI Assistant + SOC 2 + API Layer

- ☐ Build Subsurface AI Assistant (Module 9): Claude API integration with CarbonLens tool-calling. Natural language queries: “What injection rate maximises dissolution trapping for this formation?” RAG over CCUS literature corpus (18+ papers + EPA/NETL documents). CODE
- ☐ Extract API layer from GUI: expose all simulation and property endpoints as a public REST API. This unlocks Enterprise custom integration use cases and the API-as-product revenue stream that was deprioritised in v1/v2. By Year 3, the GUI has validated demand; now the API has a proven product behind it. CODE
- ☐ Complete SOC 2 Type I audit (6-month evidence collection period ends). Receive SOC 2 Type I report. Begin 12-month SOC 2 Type II observation period. SECURITY
- ☐ Hire CCUS Domain Expert (PE or Geoscientist). They lead: regulatory template accuracy verification, enterprise customer technical success, academic partnership co-authorship. HIRE

Months 29–33 Enterprise Sales Acceleration + SPE-Accredited Training

- ☐ Target 5 enterprise accounts (\$25K–100K/year): Geosyntec, TRC Environmental, Battelle, Summit Carbon, Air Products Low Carbon. Use SOC 2 Type I report in security questionnaires to unlock procurement. BIZ
- ☐ Develop SPE-accredited Continuing Professional Development (CPD) course: “CO₂ Storage Site Characterisation and Permit Preparation Using CarbonLens.” CPD accreditation is the strongest possible academic endorsement for professional adoption. ACADEMIC
- ☐ Build audit log and compliance reporting: every parameter change, simulation run, and export timestamped and attributed. Required for regulatory submissions where calculation provenance must be demonstrated. Critical for enterprise procurement. CODE
- ☐ Apply for DOE SBIR Phase II (if Phase I awarded). Target: \$1.5M–\$2M non-dilutive. Use Phase II funding to build the Subsurface AI assistant to full production quality. BIZ

Months 34–36 Decision Point: Series A or Strategic Acquisition

- ☐ Complete SOC 2 Type II audit at Month 36. SOC 2 Type II report is the final unlock for Fortune 500 and government procurement. This doubles addressable enterprise revenue. SECURITY
- ☐ Evaluate strategic acquisition offers from CMG, SLB, Halliburton Labs if presented. Decision criteria: (1) Offer must value patent portfolio + team + customer base, not just ARR multiple. (2) Retain product control for 2+ years post-acquisition. (3) Acquire only if valuation >8× ARR or \$5M minimum, whichever is higher. BIZ
- ☐ If raising Series A: target \$3–5M at \$15–20M valuation (3–4× ARR for a growing B2B SaaS with SOC 2, enterprise contracts, and patent portfolio). Use for: APAC market expansion (Southeast Asia via PETRONAS network), 3 additional engineering hires, DOE SBIR Phase II execution. BIZ

☐ **Month 36 target state:** \$1M+ ARR, 3-person team, SOC 2 Type II certified, 9-module full CCUS platform live, 8+ regulatory jurisdictions, 12+ formation presets, 2 utility patents filed, 5+ enterprise contracts, academic partnerships at 3 universities, SPE CPD course accredited. BIZ

Complete ML Model Roadmap

THE ML MODEL PORTFOLIO IS THE PRODUCT'S CORE IP

Every ML model is a defensible asset — trained on proprietary UTP lab data + literature data, producing explicit equations that are patentable, and validated against published benchmarks that engineers recognise. The full portfolio spans 9 models across the CCUS value chain. Build them in the order shown — each successive model expands the product’s coverage and the patent portfolio simultaneously.

MODEL	PROPERTY	ARCHITECTURE	PHASE	KEY PAPER BENCHMARK
Model 1 – IFT	CO ₂ –brine interfacial tension	GMDH + MGGP + MARS ensemble. Bootstrap P10/P50/P90.	PoC (Phase 0)	Beat Li et al. 2012 (R ² =0.94). Target R ² > 0.98.
Model 2 – Density	CO ₂ density (supercritical)	Span-Wagner EOS + ML correction layer (XGBoost, brine dissolution)	PoC (Phase 0)	Within 0.3% of NIST values across full T/P range
Model 3 – Viscosity	CO ₂ viscosity	Fenghour (1998) baseline + ML correction (>30 MPa)	PoC (Phase 0)	Fenghour validation dataset — must match within 2%
Model 4 – Solubility	CO ₂ solubility in brine	Duan-Sun (2003) + SVR salinity correction (>3 mol/kg)	PoC (Phase 0)	NIST Solubility Database within 2%
Model 5 – Site Scorer	Formation suitability (0–100)	Random Forest on 28 Class VI permitted sites	Phase 1	Cross-validated against EPA permitted site parameters
Model 6 – MMP	CO ₂ -EOR Minimum Miscibility Pressure	ML ensemble trained on SPE MMP literature database	Phase 1	Alston (1985) + Metcalfe (1982) benchmarks. R ² > 0.95.
Model 7 – Brine Density	CO ₂ -saturated brine density	Garcia (2001) + ML extension for mixed-salt brines	Phase 1	Garcia 2001 validation set within 0.5%
Model 8 – Diffusion	CO ₂ diffusion coefficient in brine	Literature ensemble ML (T/P/salinity)	Phase 2	Cadogan et al. 2014 experimental dataset
Model 9 – GP-UQ	Gaussian Process uncertainty quantification	Replace bootstrap with GP confidence intervals (Bell & Harvey 2021 method)	Phase 2	Calibration: sharpness + reliability on held-out test sets

Data Collection Priorities — Full Portfolio

DATASET	SOURCE	VOLUME	PHASE NEEDED
CO ₂ -brine IFT experimental	Papers A1–A5 + UTP lab measurements	1,200+ points (T, P, salinity, salt type)	Phase 0 PoC
Span-Wagner density grid	CoolProp / NIST Webbook API	50,000 synthetic points (T: 0–150°C, P: 1–100 MPa)	Phase 0 PoC
Class VI permitted site parameters	EPA Class VI Data Repository (28 sites)	28 formation parameter sets, all public	Phase 1
CO ₂ -EOR MMP literature database	SPE papers (Alston, Metcalfe, Orr groups)	500+ MMP measurements vs oil composition/T	Phase 1
CO ₂ diffusion in brine	Cadogan et al. 2014 + supplementary papers	200+ experimental measurements	Phase 2
Mixed-salt brine IFT (CaCl ₂ , MgCl ₂)	Papers A3, A6 + UTP lab expansion	300+ additional points	Phase 1–2

THE UTP LAB DATA ADVANTAGE

Every measurement added from UTP lab experiments is proprietary training data that competitors cannot replicate without running the same experiments. This data advantage compounds: 100 proprietary measurements today become the foundation for a model that performs measurably better on non-standard brine compositions than any competitor trained only on published literature data. Prioritise lab measurement expansion as a strategic investment, not just a thesis requirement.